

Technology #128

Apply silver ion (Ag+) antimicrobial coating
to the shoulder prosthesis components

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
Generated: 2026-08-30

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Apply silver ion (Ag+) antimicrobial coating to the shoulder prosthesis components

WHY DOES IT MATTER?

Documented recurring problem 'adverse_event_without_identified_device_' for shoulder prosthesis: 16 MAUDE reports across 8 distinct devices (20150515 to 20260529); root-cause evidence from 2 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 2 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 3 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, device, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, problem, producing, prosthesis, rec

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in phage_microbio. 5-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 5-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Compare infection rates between silver-coated and non-coated shoulder prostheses in a randomized controlled trial. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: CFU challenge assay against established biofilm, not only planktonic culture (measured quantity: biological response (CFU, histology)). Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use; surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation); k_{ads} is strain- and environment-specific; surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation); BIO-002 (Exponential population dynamics) is applicable only under: unlimited substrate; constant net rate over the window.; BIO-003 (Surface-limited adsorption kinetics) is applicable only under: well-mixed suspension; constant k_{ads} .; BIO-001 (Multiplicity of infection) is applicable only under: well-mixed system; planktonic or defined-surface assay.; Model input variables requiring sourced values before any numeric result: N_0 (-), r (1/s), t (s), k_{ads} (m^3/s), N_{phage} (-), $N_{bacteria}$ (-), N_{phage} (-), $N_{bacteria}$ (-) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: N (-), $dN_{infected}/dt$ (1/s), MOI (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): surface phage loading density [N_{phage} , -] value_status=UNKNOWN; infectivity retention half-life [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; host range breadth [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; moi at contact surface [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; CFU challenge load [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; resistance emergence rate [r , 1/s] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Antimicrobial coating of implants with silver to reduce periprosthetic joint infection

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'adverse_event_without_identified_device_' for shoulder prosthesis: 16 MAUDE reports across 8 distinct devices (20150515 to 20260529); root-cause evidence from 2 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 2 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 3 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, device, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, problem, producing, prosthesis, rec

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: phage_microbio

Disciplines: microbiology, surface engineering, pharmacodynamics, materials science

Architecture: PROPOSED architecture (no physical system exists): phage loading / immobilization layer; release or contact-kill mechanism; surface substrate (coating architecture); stability preservation element — realizing: Apply silver ion (Ag+) antimicrobial coating to the shoulder prosthesis components

Subsystems:

?: phage loading / immobilization layer - ?

?: release or contact-kill mechanism - ?

?: surface substrate (coating architecture) - ?

?: stability preservation element - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Apply silver ion (Ag+) antimicrobial coating to the shoulder prosthesis components

Key Physics: Microbiology / phage anti-biofilm governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Microbiology / phage anti-biofilm governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

BIO-002: Exponential population dynamics. Expression: $N = N_0 * \exp(r * t)$. Applicability to this invention: CONDITIONAL; holds under: unlimited substrate; constant net rate over the window. Assumptions: kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use. Engaged variables: NOT ESTABLISHED.

BIO-003: Surface-limited adsorption kinetics. Expression: $dN_{\text{infected}}/dt = k_{\text{ads}} * N_{\text{phage}} * N_{\text{bacteria}}$. Applicability to this invention: APPLICABLE; holds under: well-mixed suspension; constant k_{ads} . Assumptions: surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation); k_{ads} is strain- and environment-specific. Engaged variables: dN_{infected}/dt .

BIO-001: Multiplicity of infection. Expression: $MOI = N_{\text{phage}} / N_{\text{bacteria}}$. Applicability to this invention: APPLICABLE; holds under: well-mixed system; planktonic or defined-surface assay. Assumptions: surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation). Engaged variables: MOI.

Assumptions:

- kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use
- surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation)
- k_{ads} is strain- and environment-specific
- surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation)
- BIO-002 (Exponential population dynamics) is applicable only under: unlimited substrate; constant net rate over the window.
- BIO-003 (Surface-limited adsorption kinetics) is applicable only under: well-mixed suspension; constant k_{ads} .
- BIO-001 (Multiplicity of infection) is applicable only under: well-mixed system; planktonic or defined-surface assay.
- Model input variables requiring sourced values before any numeric result: N_0 (-), r (1/s), t (s), k_{ads} (m^3/s), N_{phage} (-), N_{bacteria} (-), N_{phage} (-), N_{bacteria} (-) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: N (-), dN_{infected}/dt (1/s), MOI (-) — UNKNOWN until the inputs are sourced and the model is evaluated.

- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): surface phage loading density [N_phage, -] value_status=UNKNOWN; infectivity retention half-life [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; host range breadth [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; moi at contact surface [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; CFU challenge load [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; resistance emergence rate [r, 1/s] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'adverse event without identified device ', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'infectivity loss before clinical use window', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'resistant strain emergence negating kill', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'host-range mismatch to clinical isolate', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coating wear releasing loaded phage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'biofilm matrix blocking phage access', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Devices matching 'shoulder prosthesis' must maintain function under the conditions of a recurring injury events with	SOURCE_FRONT
DI-002	Mechanism driver (proposed)	Antimicrobial coating of implants with silver to reduce periprosthetic joint infection	MODELLED
DI-003	Expected effect target	Reduction in periprosthetic joint infection rates and associated adverse events	MODELLED
DI-004	Domain design input pattern: target organ applicability NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	target organ applicability NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: device interface applicability THE DRIVERS [device]	device interface applicability THE DRIVERS [device]	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: stability applicability NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	stability applicability NOT ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate candidates found (e.g. ISO 22196 domain compliance activity) NO ENG (DEFECTIVE PROPOSED)	compliance candidate candidates found (e.g. ISO 22196 domain compliance activity) NO ENG (DEFECTIVE PROPOSED)	DEFECTIVE PROPOSED
DI-008	Custodied observation: HIP revision source https://www.prospectingarticle.com/1699660 ; robot trials https://doi.org/10.1016/j.jmb.2019.07.001	robot trials https://doi.org/10.1016/j.jmb.2019.07.001	1699660; robot trials https://doi.org/10.1016/j.jmb.2019.07.001
DI-009	Custodied observation: TDM-Guided Calvarial Implant Treatment for Complex Skull Defects Study https://doi.org/10.1016/j.jmb.2019.07.001	Study https://doi.org/10.1016/j.jmb.2019.07.001	1699660; robot trials https://doi.org/10.1016/j.jmb.2019.07.001
DI-010	Custodied observation: 3D-printed hip prosthesis https://doi.org/10.1016/j.jmb.2019.07.001 ; a state-of-the-art https://doi.org/10.1016/j.jmb.2019.07.001	a state-of-the-art https://doi.org/10.1016/j.jmb.2019.07.001	1699660; robot trials https://doi.org/10.1016/j.jmb.2019.07.001
DI-011	Custodied observation: EEG-based https://doi.org/10.1016/j.jmb.2019.07.001 ; a state-of-the-art https://doi.org/10.1016/j.jmb.2019.07.001	a state-of-the-art https://doi.org/10.1016/j.jmb.2019.07.001	1699660; robot trials https://doi.org/10.1016/j.jmb.2019.07.001
DI-012	Custodied observation: Brain-Computer https://doi.org/10.1016/j.jmb.2019.07.001 ; a state-of-the-art https://doi.org/10.1016/j.jmb.2019.07.001	a state-of-the-art https://doi.org/10.1016/j.jmb.2019.07.001	1699660; robot trials https://doi.org/10.1016/j.jmb.2019.07.001

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	phage loading / immobilization layer realizing Apply silver ion (Ag+) antimicrobial coating to the shoulder prosthesis components	CONCEPTUAL	detailed architecture/definition
DO-002	release or contact-kill mechanism realizing Apply silver ion (Ag+) antimicrobial coating to the shoulder prosthesis components	CONCEPTUAL	detailed architecture/definition
DO-003	surface substrate (coating architecture) realizing Apply silver ion (Ag+) antimicrobial coating to the shoulder prosthesis components	CONCEPTUAL	detailed architecture/definition
DO-004	stability preservation element realizing Apply silver ion (Ag+) antimicrobial coating to the shoulder prosthesis components	CONCEPTUAL	detailed architecture/definition
DO-005	phage loading / immobilization layer interfaces with release or contact-kill mechanism (interface: signal energy / material basis for Apply silver ion (Ag+) antimicrobial coating)	CONCEPTUAL	interface: signal energy / material basis for Apply silver ion (Ag+) antimicrobial coating
DO-006	release or contact-kill mechanism interfaces with surface substrate (coating architecture) (interface: signal energy / material basis for Apply silver ion (Ag+) antimicrobial coating)	CONCEPTUAL	interface: signal energy / material basis for Apply silver ion (Ag+) antimicrobial coating
DO-007	surface substrate (coating architecture) interfaces with stability preservation element (interface: signal energy / material basis for Apply silver ion (Ag+) antimicrobial coating)	CONCEPTUAL	interface: signal energy / material basis for Apply silver ion (Ag+) antimicrobial coating
DO-008	Device boundaries: patient/fluid inlet and outlet termination (control/data) prescribed by Application (flight) and antimicrobial coating to the shoulder prosthesis components	CONCEPTUAL	control/data prescribed by Application (flight) and antimicrobial coating to the shoulder prosthesis components
DO-009	Flow/force/signal path geometry realizing the mechanism of silver ion (Ag+) antimicrobial coating to the shoulder prosthesis components; every dimension is defined by the shoulder prosthesis components	UNKNOWN	3-dimensional, scaling to the shoulder prosthesis components
DO-010	Operating range for surface phage loading density within which the mechanism holds its stated behavior; computed bounds for surface phage loading density; verification of range	UNKNOWN	holds its stated behavior; computed bounds for surface phage loading density; verification of range
DO-011	Operating range for infectivity retention half-life within which the mechanism holds its stated behavior; computed bounds for infectivity retention half-life; verification of range	UNKNOWN	holds its stated behavior; computed bounds for infectivity retention half-life; verification of range
DO-012	Operating range for host range breadth within which the mechanism holds its stated behavior; computed bounds for host range breadth; verification of range	UNKNOWN	holds its stated behavior; computed bounds for host range breadth; verification of range
DO-013	Operating range for moi at contact surface within which the mechanism holds its stated behavior; computed bounds for moi at contact surface; verification of range	UNKNOWN	holds its stated behavior; computed bounds for moi at contact surface; verification of range
DO-014	Operating range for CFU challenge load within which the mechanism holds its stated behavior; computed bounds for CFU challenge load; verification of range	UNKNOWN	holds its stated behavior; computed bounds for CFU challenge load; verification of range
DO-015	Operating range for resistance emergence rate within which the mechanism holds its stated behavior; computed bounds for resistance emergence rate; verification of range	UNKNOWN	holds its stated behavior; computed bounds for resistance emergence rate; verification of range
DO-016	No closed-loop control is proposed: the invention operates as open-loop as specified; this record exists so the omission is explicit and auditable	DEFENSIVE	as specified; this record exists so the omission is explicit and auditable
DO-017	Information path: sensing element -> decision/monitor (control/data) act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical invention	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical invention
DO-018	Fixture enabling: phage infectivity retention vs time on surface (rig concept named, fixture geometry is UNKNOWN) selection; acceptance instrumentation	UNKNOWN	rig concept named, fixture geometry is UNKNOWN selection; acceptance instrumentation
DO-019	Fixture enabling: biofilm prevention CFU challenge assay (rig concept named, fixture geometry is UNKNOWN) selection; acceptance instrumentation	UNKNOWN	rig concept named, fixture geometry is UNKNOWN selection; acceptance instrumentation
DO-020	Fixture enabling: host-range and resistance-emergence rate vs time on surface (rig concept named, fixture geometry is UNKNOWN) selection; acceptance instrumentation	UNKNOWN	rig concept named, fixture geometry is UNKNOWN selection; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: CFU biofilm-prevention challenge assay	UNKNOWN	Unknown concept named, build specification, instrument selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
surface phage loading density	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
infectivity retention half-life	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
host range breadth	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
moi at contact surface	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
CFU challenge load	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
resistance emergence rate	UNKNOWN (no sourced value)	1/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
adverse event without fail-safe device	Identified adverse event without fail-safe device	design identified device	ES-Dev-15-FBD (requires design)	UNKNOWN triggered when the operating condition is
infectivity loss before sterilization	Failure of infectivity loss before sterilization	sterilization optimization	coating 2x engineering	UNKNOWN by lab tested phage particles, intended to be
resistant strain emergence	Endo by-passing kill strain emergence	phage killing bacteria	kill bacteria	UNKNOWN by lab tested phage particles, intended to be
host-range mismatch	Failure of host-range mismatch	strain isolation	phage 15-050 plate-purified	UNKNOWN whose surface characteristics differ from the
coating wear releasing phage	Failure of coating wear releasing phage	coating optimization	phage 15-050 plate-purified	UNKNOWN by lab tested phage particles, intended to be
biofilm matrix blocking phage access	Failure of biofilm matrix blocking phage access	phage access optimization	phage 15-050 plate-purified	UNKNOWN by lab tested phage particles, intended to be

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
adverse event without identification	Failure by adverse event without identification	design control: NOT ESTABLISHED	design control: NOT ESTABLISHED
infectivity loss before clinical use	Failure by infectivity loss before clinical use	stabilization/desiccation, refrigeration	High D ₁₀ values, stable at room temperature
resistant strain emergence	Failure by resistant strain emergence	negative phage control, untreated distillate, receipt inspection	NSK-001 may be phage sensitive, but not all isolates
host-range mismatch to clinical host	Failure by host-range mismatch to clinical host	isolated phage's specificity, recognition	NSK-001 strains may be specific to certain strains
coating wear releasing loaded phage	Failure by coating wear releasing loaded phage	phage resistance, minimal abrasion, coating	NSK-001 phage resistance to abrasion
biofilm matrix blocking phage access	Failure by biofilm matrix blocking phage access	degrading of matrix, penetration	NSK-001 phage penetration of biofilm

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Compare infection rates between silver-coated and uncoated biofilms	Compare infection rates between silver-coated and uncoated biofilms	Compare infection rates between silver-coated and uncoated biofilms	Compare infection rates between silver-coated and uncoated biofilms
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	CFU challenge assay against established biofilm	CFU challenge assay against established biofilm	CFU challenge assay against established biofilm	CFU challenge assay against established biofilm

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed	NOT ESTABLISHED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	electrospun + metal-coated catheter	semi-industrial device coatings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
electrospun + metal-coated catheter	domain candidate material	1	Material	electrospun + metal-coated catheter	semi-industrial	High	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- electrospinning + immobilization dip under aseptic control: domain registry manufacturing route - candidate only; no process qualification exists
- lyophilization/storage stabilization: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-001', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter surface phage loading density [N_phage -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter infectivity retention half-life [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter host range breadth [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter moi at contact surface [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter CFU challenge load [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}

- {'unknown': 'value of critical parameter resistance emergence rate [r 1/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	measuring the prop...	physical verification (0.5-1000 Hz) ...	ENGINEERING_PROPOSED (no sourced value exists)	first physical article	100 hours (1 person)
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	NOT ESTABLISHED (constraint compliance)	constraint-compliance envelope article	100 hours (1 person)
WP-03	article for VF-001	per method	Compare infection rates of 5000 and 10000	ENGINEERING_PROPOSED (no sourced value exists)	article for VF-001	100 hours (1 person)
WP-04	article for VF-002	per method	run killer experiment: 10000 and 5000	NOT ESTABLISHED (no sourced value exists)	article for VF-002	100 hours (1 person)
WP-05	article for VF-003	per method	CFU challenge assay	NOT ESTABLISHED (no sourced value exists)	article for VF-003	100 hours (1 person)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 5-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.